

RCP Future Input Processing

Setup

```
packages <- c("readxl", "dplyr", "ISOweek")
invisible(lapply(packages, library, character.only = TRUE))

find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) return(path)
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}

base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()

rcp_dir <- file.path(base_dir, "Data/Simulation_data/UTCI")
scenarios <- c("rcp26", "rcp85")
regions <- c("Attiki", "Lisbon", "Roma")
```

Helper Functions

```
fill_na_linear <- function(x) {
  x <- as.numeric(x)
  idx <- which(is.finite(x))
  if (length(idx) <= 1) return(x)
  approx(idx, x[idx], xout = seq_along(x), rule = 2)$y
}

fill_na_nearest <- function(x) {
  x <- as.numeric(x)
  if (length(x) == 0 || all(!is.finite(x))) return(x)

  for (i in seq_along(x)) {
    if (!is.finite(x[i]) && i > 1) x[i] <- x[i - 1]
  }
  for (i in rev(seq_along(x))) {
    if (!is.finite(x[i]) && i < length(x)) x[i] <- x[i + 1]
  }
  x
}
```

```

read_rcp_utci <- function(scenario, type) {
  path <- file.path(rcp_dir, scenario, paste0("UTCI_", type, "_", scenario, ".xlsx"))
  data <- read_xlsx(path, .name_repair = "minimal")[, -1, drop = FALSE]

  date_col <- intersect(c("Time", "Date"), names(data))[1]
  if (is.na(date_col)) stop("Cannot find Time or Date column in ", path)

  value_cols <- setdiff(names(data), date_col)
  for (col in value_cols) {
    if (is.numeric(data[[col]])) data[[col]] <- fill_na_linear(data[[col]])
  }

  data$Week <- ISOweek(as.Date(data[[date_col]]))
  as.data.frame(data)
}

classify_utci <- function(x) {
  case_when(
    x < -13 ~ -1,
    x <= 32 ~ 0,
    x > 32 ~ 1,
    TRUE ~ NA_real_
  )
}

make_wave_list <- function(max_data, min_data) {
  wave_list <- lapply(regions, function(region) {
    hot_level <- classify_utci(max_data[[region]])
    cold_level <- classify_utci(min_data[[region]])

    hot <- as.integer(hot_level == 1 &
                      lag(hot_level, 1) == 1 &
                      lag(hot_level, 2) == 1)
    cold <- as.integer(cold_level == -1 &
                      lag(cold_level, 1) == -1 &
                      lag(cold_level, 2) == -1)

    hot[is.na(hot)] <- 0
    cold[is.na(cold)] <- 0

    hot_week <- rowsum(hot, max_data$Week, reorder = FALSE)
    cold_week <- rowsum(cold, min_data$Week, reorder = FALSE)
    week <- intersect(rownames(hot_week), rownames(cold_week))

    out <- data.frame(
      hot_wave = as.numeric(hot_week[week, 1]),
      cold_wave = as.numeric(cold_week[week, 1])
    )
    colnames(out) <- c(paste0(region, "_hot_wave3"),
                      paste0(region, "_cold_wave3"))
    rownames(out) <- week
    out
  })
}

```

```

names(wave_list) <- c("Attiki", "Lisbon", "Rome")
wave_list
}

make_lag21_matrix <- function(mean_data, region, start_day = 7) {
  start <- switch(
    as.character(start_day),
    "1" = c(1, 7, 6, 5, 4, 3, 2),
    "4" = c(4, 3, 2, 1, 7, 6, 5),
    "7" = c(7, 6, 5, 4, 3, 2, 1),
    stop("start_day must be 1, 4, or 7")
  )

  weekly <- mean_data %>%
    select(Week, value = all_of(region)) %>%
    group_by(Week) %>%
    summarise(value = list(value), .groups = "drop")

  out <- matrix(NA_real_, nrow = nrow(weekly), ncol = 22)
  colnames(out) <- paste0("lag", 0:21)
  rownames(out) <- weekly$Week

  get_day <- function(day) {
    vapply(weekly$value, function(x) {
      if (length(x) >= day) as.numeric(x[[day]]) else NA_real_
    }, numeric(1))
  }

  for (i in seq_len(7)) {
    day_value <- get_day(start[i])
    out[, i] <- day_value
    out[, i + 7] <- lag(day_value, 1)
    out[, i + 14] <- lag(day_value, 2)
  }
  out[, 22] <- lag(get_day(start[1]), 3)

  out <- apply(out, 2, fill_na_nearest)
  colnames(out) <- paste0("lag", 0:21)
  rownames(out) <- weekly$Week
  out
}

```

Process RCP Scenarios

```

generated_files <- list()

for (scenario in scenarios) {
  scenario_dir <- file.path(rcp_dir, scenario)

  rcp_mean <- read_rcp_utci(scenario, "mean")
  rcp_min <- read_rcp_utci(scenario, "min")
  rcp_max <- read_rcp_utci(scenario, "max")
}

```

```

wave_list <- make_wave_list(rcp_max, rcp_min)
wave_object <- paste0("wave_list_", scenario)
wave_file <- file.path(scenario_dir, paste0(wave_object, ".RData"))
assign(wave_object, wave_list)
save(list = wave_object, file = wave_file)
generated_files[[length(generated_files) + 1]] <- data.frame(
  scenario = scenario,
  file = basename(wave_file)
)

for (region in regions) {
  cb_matrix <- make_lag21_matrix(rcp_mean, region, start_day = 7)
  cb_object <- paste0(region, "_prep_cb_", scenario)
  cb_file <- file.path(scenario_dir, paste0(region, "_cb_", scenario, ".RData"))
  assign(cb_object, cb_matrix)
  save(list = cb_object, file = cb_file)
  generated_files[[length(generated_files) + 1]] <- data.frame(
    scenario = scenario,
    file = basename(cb_file)
  )
}
}

generated_files <- do.call(rbind, generated_files)
data.frame(
  scenario = names(table(generated_files$scenario)),
  n_files = as.integer(table(generated_files$scenario)),
  row.names = NULL
)

##   scenario n_files
## 1    rcp26      4
## 2    rcp85      4

```